Field	Details
Student	Makhaola Nkhahle, Student no. 202403142, GitHub: MAKHAOLA-N
Team	Binary Brains (lab group A, B, C)
Role	ISA lead: overview, register file
Delivered	Pull request #1 (ISA overview and register file)
Repository	https://github.com/MAKHAOLA-N/LIMO_BINARY_BRAINS (release tag m1)

1. ISA Overview

LIMO is a 32-bit load-store instruction set with Sesotho mnemonics, derived from RV32I. It has 15 instructions in four formats (R, I, S, B). The table below states each design choice and the reason for it.

Specification	Rationale
32-bit load-store; fixed 32-bit instructions; PC advances by 4 bytes.	Fixed width simplifies fetch and decode; instructions are word-aligned.
16 x 32-bit registers; 4-bit indices.	Saves one bit per register operand versus RV32I; increases register pressure.
Register, immediate, base + byte offset, and PC-relative branch addressing.	Supports arithmetic/logic, word memory access and conditional branches.
Byte-addressed, little-endian memory; word accesses; aligned instructions.	Simple memory model. Misaligned data addresses are rounded down in this subset.
Formats R, I, S, B; no U or J.	No upper-immediate or jump instruction is included in this 15-instruction subset.
No multiply/divide, floating point, CSR or architectural exceptions.	Limits implementation complexity for the milestone.

2. Register File

Sixteen 32-bit registers use 4-bit indexes. Register r0 is hard-wired to zero. The PC is separate architectural state and is not encoded in register fields.

Number	Sesotho name / alias	Type	Purpose and convention
0	r0 / lefela	SPR	Constant zero; writes ignored.
1-13	r1-r13 / n1-n13	GPR	General-purpose registers.
14	r14 / mokokotlo (sp)	SPR	Stack pointer.
15	r15 / aterese ea khutlo (ra)	SPR	Return address.
PC	khaontara ea lenaneo	SPR	Byte address of the current instruction; dedicated, not encoded.

3. Design Rationale

Register-file size. Sixteen 32-bit registers use 4-bit indexes instead of the 5-bit indexes of RV32I. An R-type instruction with three register operands saves three encoding bits, which are used to provide wider function fields. Forwarding comparators also narrow from 5 to 4 bits. The file holds 16 x 32 = 512 bits. Because r0 is constant zero and r14/r15 are reserved for the stack pointer and return address, 13 general-purpose registers remain.